dependence a VAR-based network is meant to capture; weekly aggregation mitigates the non-synchronous trading between Asian and EMEA local markets and their US-listed ETFs; and it reduces the microstructure noise of daily ETF prices. We also choose to study returns rather than volatilities. Diebold and Yılmaz (2014) measure the connectedness of realized volatilities; with daily prices the same could be done here, by building weekly realized volatilities, but we leave volatility connectedness as an extension and focus on the transmission of return shocks. Returns are not devoid of cross-market predictive content: in a full-sample VAR(1) on the 18 series, 59 of the 324 slope coefficients (18.2%) are significant at the 5% level, against 5% expected by chance. Our networks therefore describe the transmission of return shocks, not of risk.

Descriptive statistics. Table 1 reports summary statistics. All series are strongly leptokurtic and mostly negatively skewed; an augmented Dickey–Fuller test rejects a unit root for every series. Five weekly log returns exceed 25% in absolute value (ARGT in the week of the August 2019 primary elections and four observations in March 2020); they are genuine market moves and are kept. The March 2020 crash dominates the tails: of the ten largest weekly drops pooled across the 17 emerging markets, eight occur in the weeks ending 13 and 20 March 2020, and 14 of the 17 markets record their worst week in March 2020.

Table 1. Descriptive statistics of weekly log returns, 2011-03-11 to 2026-10-23 (816 weeks). Kurtosis is not in excess.

Ticker	Country	Region	Start	Mean (%)	Std (%)	Min (%)	Max (%)	Skew.	Kurt.
ARGT	Argentina	Latin America	2011-03-11	0.15	4.18	-28.4	24.0	-0.51	8.9
ECH	Chile	Latin America	2011-03-11	-0.03	3.56	-21.2	16.2	-0.36	5.6
EWZ	Brazil	Latin America	2011-03-11	0.01	4.26	-26.5	22.2	-0.18	6.4
EWW	Mexico	Latin America	2011-03-11	0.07	3.29	-24.0	10.3	-0.74	7.8
EPU	Peru	Latin America	2011-03-11	0.12	3.35	-19.3	13.3	-0.39	5.9
EWY	South Korea	Asia	2011-03-11	0.14	3.66	-20.3	16.0	-0.40	6.6
EWT	Taiwan	Asia	2011-03-11	0.23	2.85	-12.6	11.2	-0.43	4.4
EPI	India	Asia	2011-03-11	0.09	2.90	-18.8	13.3	-0.35	6.3
FXI	China	Asia	2011-03-11	0.01	3.47	-12.6	17.0	0.03	4.5
THD	Thailand	Asia	2011-03-11	0.07	2.91	-23.5	9.6	-0.85	9.0
EIDO	Indonesia	Asia	2011-03-11	-0.08	3.68	-33.6	20.7	-1.25	14.6
EWM	Malaysia	Asia	2011-03-11	0.01	2.25	-12.2	10.9	-0.22	6.5
VNM	Vietnam	Asia	2011-03-11	-0.01	3.15	-19.3	11.9	-0.35	5.9
EPHE	Philippines	Asia	2011-03-11	0.02	2.92	-19.3	14.4	-0.57	7.4
EZA	South Africa	EMEA	2011-03-11	0.07	4.06	-26.9	14.1	-0.57	6.5
TUR	Turkey	EMEA	2011-03-11	-0.03	4.60	-23.0	20.0	-0.63	6.1
EPOL	Poland	EMEA	2011-03-11	0.08	3.66	-25.1	15.2	-0.58	6.5
SPY	United States	Global	2011-03-11	0.25	2.23	-15.7	11.4	-0.74	9.3

Limitations. (i) ETF returns embed exchange-rate movements against the USD and are subject to non-synchronous trading. (ii) FXI tracks the FTSE China 50 Index, i.e. 50 large-cap Chinese companies listed in Hong Kong, and EPI tracks the earnings-weighted WisdomTree India Earnings Index, so neither is a broad market index of its country. (iii) Colombia is absent. (iv) Liquidity requirements and the common start date exclude younger funds. (v) Volatility connectedness, the object of the base paper, is not studied here.

4 Methods

Let $y_t \in \mathbb{R}^N$ be the vector of weekly returns ($N = 18$). All methods are estimated on rolling windows of $w = 104$ weeks; each estimate is dated at the last week of its window. For the comparison, all methods use the same 179 windows (step of four weeks, first window ending 1 March 2013).

4.1 Diebold–Yilmaz connectedness

We fit a VAR(p), $y_t = c + \sum_{k=1}^{p} A_k y_{t-k} + \varepsilon_t$, $\mathbb{E}[\varepsilon_t \varepsilon_t'] = \Sigma$, by OLS, with $p \in \{1, 2\}$ selected by BIC in each window. The cap $p \leq 2$ is an adaptation: with $N = 18$ and about 100 observations per window, $p = 3$ would mean 55 regressors per equation. With MA coefficients Ψ_h ($\Psi_0 = I$), the H-step generalized variance share (Pesaran and Shin, 1998) is

$$\theta_{ij}(H) = \frac{\sigma_{jj}^{-1} \sum_{h=0}^{H-1} (e_i' \Psi_h \Sigma e_j)^2}{\sum_{h=0}^{H-1} e_i' \Psi_h \Sigma \Psi_h' e_i}, \qquad \tilde{\theta}_{ij} = \theta_{ij} \Big/ \sum_j \theta_{ij}, \tag{1}$$

which is invariant to the ordering of the variables. Following Diebold and Yılmaz (2014), $C_i^{\text{from}} = 100 \sum_{j \neq i} \tilde{\theta}_{ij}$, $C_j^{\text{to}} = 100 \sum_{i \neq j} \tilde{\theta}_{ij}$, net connectedness $C_i^{\text{net}} = C_i^{\text{to}} - C_i^{\text{from}}$, pairwise net $100(\tilde{\theta}_{ji} - \tilde{\theta}_{ij})$, and total connectedness $C = \frac{100}{N} \sum_{i \neq j} \tilde{\theta}_{ij}$. The baseline horizon is $H = 4$ weeks. The base paper uses daily data, 100-day windows and $H = 12$ days; we keep the logic (a horizon of about one month, a window long enough for a stable VAR) on the weekly scale.

4.2 Sparse VAR (lasso)

As a bridge between the dense DY VAR and the sparse network estimators, we estimate each VAR equation by the lasso (Tibshirani, 1996) on lags $1, \ldots, 4$ (standardized regressors, unpenalized intercept), so that the penalty performs lag selection. The penalty is chosen per equation by time-series cross-validation (five expanding folds, test blocks always after the training block; random K-fold would leak future information), over 40 log-spaced values; the extended BIC (Chen and Chen, 2008) is an alternative. The same GFEVD and connectedness measures are computed on the lasso VAR, so differences with DY isolate the effect of regularization.

4.3 NETS

We estimate NETS with one lag, i.e. a sparse VAR(1) and a sparse concentration matrix $K = \Sigma^{-1}$ of the innovations, using adaptive lasso weights (Zou, 2006; Peng et al., 2009). Since the `nets` package requires the penalty to be supplied, we select it by BIC over 15 log-spaced values. The Granger network is given by the non-zero entries of A_1, the contemporaneous network by the partial correlations $-K_{ij}/\sqrt{K_{ii}K_{jj}}$, and the long-run network by the partial correlations of $K_{LR} = (I - A(1))' K (I - A(1))$. Each window is standardized before estimation.

4.4 FNETS

Writing $y_t = \chi_t + \xi_t$ for the common and idiosyncratic components, we use the unrestricted (generalized dynamic) factor model, select q with the Hallin and Liška (2007) information criterion (criterion IC5, the package default), estimate the idiosyncratic VAR(1) by lasso with the penalty chosen by the package's BIC, and estimate contemporaneous and long-run partial correlations of ξ_t with the package's CLIME-type estimator (Cai et al., 2016) and adaptive thresholding. The share of variance explained by the common component is $\text{var}(\chi_{it})/\text{var}(y_{it})$. Applying the GFEVD to the idiosyncratic VAR yields *idiosyncratic connectedness*; comparing it with DY measures how much connectedness is due to the common factor. With $N = 18$ and $T = 104$ the estimated idiosyncratic innovation covariance is often not positive definite; we project it onto the positive-definite cone by clipping eigenvalues at 1% of the mean absolute eigenvalue, and clip partial correlations to $[-1, 1]$ (the support of the network is unchanged). Both adjustments, and the fact that the idiosyncratic partial-correlation network can be empty in long samples (Section 5), indicate that FNETS operates here at the edge of its intended setting of large N and T; we treat its idiosyncratic networks with corresponding caution.

4.5 Comparable measures and comparison criteria

Every method is mapped to the same objects: a VAR on which the same GFEVD connectedness is computed; a directed Granger network (sum over lags of the absolute coefficients); and contemporaneous and long-run partial-correlation networks (for DY and the lasso VAR, implied by the estimated VAR). We compare methods on three criteria:

1. *Density*: share of non-zero links in each network; for the dense GFEVD table, share of pairwise shares above $1/N$; and total connectedness.

2. *Agreement*: correlation of total and net connectedness series across methods; per-window Spearman correlation between the off-diagonal entries of two methods' matrices; Jaccard similarity of their strongest 10% of links.

3. *Stability*: variability of total connectedness, and Spearman correlation and top-link Jaccard similarity between consecutive windows' matrices.

Finally, the share of DY connectedness that disappears once the common factor is removed is $1 - C^{\text{FNETS}}/C^{\text{DY}}$, averaged over windows.

4.6 Robustness

We rerun all methods with windows of 78 and 130 weeks, horizons of 2 and 8 weeks, without SPY, and excluding 2020. For the last, every rolling window containing any week of 2020 is dropped (the resulting gap is not interpolated), and static full-period connectedness is estimated separately for 2011–2019 and 2021–2026.

5 Results

5.1 Full-sample networks

Table 4 (appendix) reports static connectedness estimated once on the whole sample; these figures differ from the averages over rolling windows reported in Table 2 and below (e.g. 81.9% against 81.0% for Diebold–Yilmaz, 34.2% against 47.2% for FNETS), because a single long-sample estimate is not the mean of short-window estimates. The two dense estimators give almost the same total connectedness (Diebold–Yilmaz 81.9%, lasso VAR 82.1%), NETS gives 73.5%, and the idiosyncratic component identified by FNETS gives only 34.2%. The heatmap of Figure 3(a) shows that Diebold–Yilmaz connectedness is spread widely across all pairs, with only mild regional clustering: the strongest pairs are Korea–Taiwan and the Indonesia–Malaysia–Philippines–Thailand block, and South Africa is a common source of shocks for almost every market. Once the common factor is removed (Figure 3(b)), the remaining pairwise shares are concentrated in a few pairs (Korea–Taiwan above all), and the full-sample idiosyncratic partial-correlation network is empty (Figure 4, right panel). This can mean either that, once the common factor is removed, there is little network structure left among these markets, or that the thresholded estimator is too strict to detect it with the available data; in rolling 104-week windows the idiosyncratic network is sparse (7% of possible long-run links on average) but not empty. We cannot distinguish the two explanations and list this among the limitations, together with the positive-definite adjustment described in Section 4.4. NETS retains 92 long-run partial-correlation links (60% of all possible pairs). The lagged (Granger) dependence is very weak in all sparse estimators: lasso, NETS and FNETS keep only 2.5%, 1.7% and 0.3% of the possible directed lag links in the rolling windows, so connectedness at the weekly frequency is overwhelmingly contemporaneous.